Applied Data Science in FinTech

1. COURSE SPECIFICATIONS

COURSE TITLE	Applied Data Science in Fintech
TEACHING LANGUAGE	English
COURSE SUPERVISOR	Dr. Mario Gellrich Zurich University of Applied Sciences, Switzerland
OFFICE HOURS	On appointment (at school or by MS Teams)
CLASSROOM(S)	See course portal
COURSE HOURS	2 x 6 hours lectures and exercises
FACILITATORS	

Data science combines computational, statistical, and algorithmic methods to analyze data and generate insights. In this course, these methods are explored through Python-based agent-based modeling, simulation analysis, credit risk modeling, and interactive web applications for teaching and experimentation.

The course takes place over two consecutive days. On the first day, students are introduced to cellular automata and agent-based modeling in Python. Students will learn how local interaction rules can generate complex system-level behavior and how to build, run, and analyze simulation models in interactive and notebook-based environments. On the second day, the focus shifts to credit risk, where students develop and explore an agent-based simulation model for credit risk assessment. Across the course, the exercises cover core stages of practical data science, including problem framing, model development, simulation-based analysis, sensitivity analysis, visualization, and the deployment of models and algorithms in interactive applications.

2. LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Frame real-world problems as agent-based models, including credit risk, segregation, wealth distribution, and ecological dynamics.
- Use Python to build, run, and analyze simulation models in interactive applications.
- Develop multi-agent models to understand how individual behavior and local interactions create system-level outcomes.
- Apply sensitivity analysis, metrics, charts, and heatmaps to compare scenarios and interpret simulation results.
- Deploy models and algorithms as interactive web applications for experimentation, communication, and decision

3. EVALUATION

Assessment activities:

Participants will attend lectures and perform practical lab exercises with the Python programming language. Hence, participants will have a good mixture of theory and hands-on exercises.

Evaluation assignments and criteria:

Student evaluations will be accessed around the following deliverables:

- Implementation and analysis of simulation models in Python, including agent-based modeling, and credit risk applications.
- Development and adaptation of an interactive modeling application or analytical workflow based on the course materials.
- Presentation and interpretation of results, including model behavior, parameter effects, sensitivity analysis, visualizations, and key findings.

4. LEARNING ACTIVITIES

Organization, methods and pedagogy:

- Workshops
- Individual and group exercises
- Presentations

Bibliography:

Materials will be updated regularly on the course portal at http://baisummer.com.